

# Plant Leaf Disease Classification Report

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## 1. Datasets Overview

A summary of the specific datasets used for training and validation across the experiments.

| Dataset Name                   | Total Size (Records) | Classes |
|--------------------------------|----------------------|---------|
| PlantVillage                   | ~54,303              | 38      |
| Wheat Plant Diseases           | 14,155               | 15      |
| Rice Disease Dataset           | 45,400               | 19      |
| Plant-Doc                      | 2,572                | 28      |
| Potato and Tomato Disease Data | 842 (Test Set)       | 10      |

## 2. Models Overview

### CNN Architectures:

- **EfficientNet B3:** A highly efficient convolutional neural network utilizing compound scaling, which uniformly scales all dimensions of depth, width, and resolution using a single parameter.

### Segmentation Pipelines:

To force models to isolate diseased leaf areas and ignore background noise, a progressive instance segmentation approach was adopted using the Segment Anything Model (SAM).

- **SAM 1 (Base, Large, Huge):** The first generation of the promptable segmentation model. Built on three Vision Transformer backbones:
  - **SAM 1 Base :** The fastest and most lightweight variant.
  - **SAM 1 Large :** The middle-tier variant balancing speed and prediction quality.
  - **SAM 1 Huge :** The heaviest variant providing the most detailed original mask predictions.These models were adapted in this project to utilize bounding box prompting to crop out the leaves.
- **SAM 2 (Base, Large, Huge):** The second generation model featuring a hierarchical architecture designed for more complex masking tasks.
- **SAM 3:** The latest iteration, provided as a single, highly refined unified version. Utilized as an aggressive preprocessing pipeline to cleanly extract leaf segments, allowing for the experimental mixing of different background configurations (e.g., Grayscale vs. Segmented data) before classification.

### Vision Transformers (ViTs):

State of the art transformer architectures leveraging self attention mechanisms and self supervised learning to organically extract features.

- **DINOv2 (Small, Base, Large):** The second generation visual foundation model, trained on 142 million diverse images using self supervised distillation.
  - **DINOv2 Small:** A highly efficient ~22 million parameter model.
  - **DINOv2 Base :** An ~86 million parameter architecture providing stable feature extraction capable of adapting to complex, multi crop domains.

- **DINOv2 Large:** A heavy ~300 million parameter variant providing deeper feature extraction.
- **DINOv3 (Small, Base, Large):** The third generation architecture, distilled from a massive 7 billion parameter model across 1.7 billion unannotated images.
  - **DINOv3 Small :** A lightweight variant designed for faster inference and lower resource environments.
  - **DINOv3 Base :** A mid weight architecture providing the optimal balance between feature extraction depth and computational efficiency.
  - **DINOv3 Large :** A massive architecture. Its sheer parameter capacity is designed to capture highly complex, structural feature embeddings that are resilient to severe data degradation (such as grayscale conversion), reducing the architectural reliance on external SAM preprocessing.

### 3. Best Performing Models by Dataset

The following table highlights the absolute best performing architecture for each distinct dataset/subset tested during this project, including their detailed classification metrics.

| Dataset / Subset                     | Best Model                      | Validation Accuracy | Testing Accuracy | Precision (Weighted) | F1-Score (Weighted) |
|--------------------------------------|---------------------------------|---------------------|------------------|----------------------|---------------------|
| Rice Disease Dataset (19 Classes)    | DINOv2 Base (4 Blocks Unfrozen) | 93.83%              | <b>94.62%</b>    | 0.95                 | 0.95                |
| PlantVillage + PlantDoc (10 Classes) | DINOv2 Base (2 Blocks Unfrozen) | 91.44%              | <b>92.09%</b>    | 0.92                 | 0.92                |
| Wheat Plant Diseases (15 Classes)    | DINOv2 Base (2 Blocks Unfrozen) | 92.86%              | <b>91.73%</b>    | 0.94                 | 0.90                |
| Potato + Tomato (10 Classes)         | DINOv2 Base (4 Blocks Unfrozen) | 69.37%              | <b>70.67%</b>    | 0.71                 | 0.70                |

### 4. Comprehensive Results Tables

The following tables document the complete testing results of the project, detailing every model, dataset configuration, architectural tweak, and the resulting validation and testing metrics.

#### 4.1 Early Baselines & Segmentation Pipelines (CNNs & SAM)

Initial experiments focused on standard CNNs and the progression of the Segment Anything Model (SAM) to isolate leaf features and combat background memorization.

| Model / Pipeline | Preprocessing / Dataset Used | Classes | Validation Accuracy | Testing Accuracy |
|------------------|------------------------------|---------|---------------------|------------------|
| EfficientNet B3  | PlantVillage + PlantDoc      | All     | 99.67%              | 25.68%           |
| EfficientNet B0  | PlantVillage + PlantDoc      | All     | 96.00%              | 49.00%           |
| SAM 1 Pipeline   | Normal PV + Seg PD           | 4       | 85.19%              | 58.00%           |
| SAM 2 Base       | Normal PV + Seg PD           | 4       | 74.00%              | 69.00%           |
| SAM 2 Large      | Normal PV + Seg PD           | 4       | 70.00%              | 59.00%           |
| SAM 2 Base       | Normal PV + Seg PD           | 13      | 55.00%              | 51.00%           |

|                       |                              |          |               |               |
|-----------------------|------------------------------|----------|---------------|---------------|
| SAM 3 Pipeline        | Normal PV + Seg PD           | 4        | 76.00%        | 69.23%        |
| SAM 3 Pipeline        | Segmented PV + Seg PD        | 5        | 86.00%        | 82.60%        |
| SAM 3 Pipeline        | Normal PV + Seg PD           | 5        | 87.15%        | 78.26%        |
| SAM 3 Pipeline        | Grayscale PV + Seg PD        | 5        | 86.00%        | 73.91%        |
| SAM 3 Pipeline        | Segmented Datasets           | 5        | 86.00%        | 83.66%        |
| SAM 3 Pipeline        | Segmented PV + Seg PD        | 5        | 82.23%        | 87.75%        |
| <b>SAM 3 Pipeline</b> | <b>Grayscale PV + Seg PD</b> | <b>5</b> | <b>86.00%</b> | <b>89.79%</b> |
| SAM 3 Pipeline        | Segmented PV + Seg PD        | 10       | 83.42%        | 81.00%        |
| SAM 3 Pipeline        | Normal PV + Seg PD           | 10       | 85.56%        | 81.00%        |
| SAM 3 Pipeline        | Grayscale PV + Seg PD        | 10       | 86.63%        | 85.26%        |

#### 4.2 DINOv2 Architecture Evaluations (10 Class PlantVillage + PlantDoc)

Experiments utilizing the second generation Vision Transformer. The goal was to find the optimal layer unfreezing depth to bridge the domain gap between lab and real world images.

| Model Size         | Preprocessing / Dataset Used | Configuration / Backbone     | Validation Accuracy | Testing Accuracy |
|--------------------|------------------------------|------------------------------|---------------------|------------------|
| DINOv2 Small       | Grayscale PV + Seg PD        | Fully Frozen (224x224)       | 80.75%              | 70.53%           |
| DINOv2 Small       | Normal PV + Normal PD        | Fully Frozen (224x224)       | 85.03%              | 72.63%           |
| DINOv2 Small       | Segmented PV + Seg PD        | Fully Frozen (224x224)       | 80.21%              | 76.84%           |
| DINOv2 Small       | Segmented PV + Seg PD        | Unfroze 2 Blocks (Weighted)  | 86.10%              | 83.16%           |
| DINOv2 Small       | Segmented PV + Seg PD        | Unfroze 4 Blocks (256→224)   | 89.30%              | 88.42%           |
| DINOv2 Small       | Segmented PV + Seg PD        | Unfroze 2 Blocks (5-Fold CV) | ~98.35%             | 90.50%           |
| DINOv2 Small       | Segmented PV + Seg PD        | Unfroze 2 Blocks (256→224)   | 89.30%              | 90.53%           |
| DINOv2 Base        | Grayscale PV + Seg PD        | Unfroze 2 Blocks             | 87.70%              | 87.37%           |
| DINOv2 Base        | Segmented PV + Seg PD        | Unfroze 2 Blocks             | 88.24%              | 89.47%           |
| DINOv2 Base        | Normal PV + Seg PD           | Unfroze 2 Blocks             | 86.10%              | 90.40%           |
| DINOv2 Base        | Segmented PV + Seg PD        | Unfroze 2 Blocks (Expanded)  | 88.24%              | 91.53%           |
| <b>DINOv2 Base</b> | <b>Normal PV + Normal PD</b> | <b>Unfroze 2 Blocks</b>      | <b>91.44%</b>       | <b>92.09%</b>    |
| DINOv2 Large       | Normal PV + Normal PD        | Unfroze 2 Blocks             | 89.84%              | 83.16%           |
| DINOv2 Large       | Normal PV + Normal PD        | Unfroze 3 Blocks             | 90.91%              | 84.21%           |
| DINOv2 Large       | Grayscale PV + Seg PD        | Unfroze 3 Blocks             | 88.24%              | 88.70%           |
| DINOv2 Large       | Segmented PV + Seg PD        | Unfroze 3 Blocks             | 88.24%              | 88.70%           |

|              |                       |                                     |         |        |
|--------------|-----------------------|-------------------------------------|---------|--------|
| DINOv2 Large | Normal PV + Normal PD | Unfroze 3 Blocks (+ Regularization) | 90.91%  | 89.27% |
| DINOv2 Large | Normal PV + Normal PD | Unfroze 4 Blocks                    | 91.98%  | 89.27% |
| DINOv2 Large | Normal PV + Normal PD | Unfroze 6 Blocks (384x384)          | ~92.00% | 89.83% |

### 4.3 DINOv3 Architecture Evaluations (10 Class PlantVillage + PlantDoc)

Testing the state of the art third generation Vision Transformer to evaluate its innate ability to organically ignore chaotic backgrounds without SAM preprocessing.

| Model Size          | Preprocessing / Dataset Used | Configuration / Backbone | Validation Accuracy | Testing Accuracy |
|---------------------|------------------------------|--------------------------|---------------------|------------------|
| DINOv3 Small        | Normal PV + Seg PD           | Unfroze 2 Blocks         | 84.49%              | 85.88%           |
| DINOv3 Small        | Segmented PV + Seg PD        | Unfroze 2 Blocks         | 83.96%              | 87.01%           |
| DINOv3 Small        | Normal PV + Normal PD        | Unfroze 2 Blocks         | 90.37%              | 87.57%           |
| DINOv3 Small        | Normal PV + Normal PD        | Unfroze 6 Blocks         | 90.37%              | 87.01%           |
| DINOv3 Small        | Normal PV + Normal PD        | Unfroze 4 Blocks         | 90.91%              | 89.27%           |
| DINOv3 Base         | Grayscale PV + Seg PD        | Unfroze 6 Blocks         | 82.35%              | 84.18%           |
| DINOv3 Base         | Normal PV + Normal PD        | Unfroze 2 Blocks         | 93.05%              | 88.14%           |
| DINOv3 Base         | Segmented PV + Seg PD        | Unfroze 6 Blocks         | 88.00%              | 90.00%           |
| DINOv3 Base         | Normal PV + Seg PD           | Unfroze 6 Blocks         | 82.89%              | 90.40%           |
| <b>DINOv3 Base</b>  | <b>Normal PV + Normal PD</b> | <b>Unfroze 6 Blocks</b>  | <b>90.37%</b>       | <b>90.96%</b>    |
| DINOv3 Large        | Segmented PV + Seg PD        | Unfroze 4 Blocks         | 86.00%              | 89.00%           |
| DINOv3 Large        | Normal PV + Normal PD        | Unfroze 4 Blocks         | 91.44%              | 89.27%           |
| DINOv3 Large        | Grayscale PV + Seg PD        | Unfroze 4 Blocks         | 90.91%              | 90.40%           |
| <b>DINOv3 Large</b> | <b>Normal PV + Seg PD</b>    | <b>Unfroze 4 Blocks</b>  | <b>89.84%</b>       | <b>90.96%</b>    |

### 4.4 Dataset Specific Generalization Tests (DINOv2 Base)

Evaluations applying the highly stabilized DINOv2 Base architecture to alternative crop datasets to measure broad agricultural generalization.

| Dataset Tested                          | Preprocessing             | Configuration / Backbone | Validation Accuracy | Testing Accuracy |
|-----------------------------------------|---------------------------|--------------------------|---------------------|------------------|
| Potato & Tomato (10 Classes)            | Normal Unsegmented        | Unfroze 2 Blocks         | 69.52%              | 69.24%           |
| <b>Potato &amp; Tomato (10 Classes)</b> | <b>Normal Unsegmented</b> | <b>Unfroze 4 Blocks</b>  | <b>69.37%</b>       | <b>70.67%</b>    |
| Wheat Disease (15 Classes)              | Normal Unsegmented        | Unfroze 4 Blocks         | 92.14%              | 91.60%           |

|                            |                    |                  |        |        |
|----------------------------|--------------------|------------------|--------|--------|
| Wheat Disease (15 Classes) | Normal Unsegmented | Unfroze 2 Blocks | 92.86% | 91.73% |
| Rice Disease (19 Classes)  | Normal Unsegmented | Unfroze 2 Blocks | 93.30% | 94.19% |
| Rice Disease (19 Classes)  | Normal Unsegmented | Unfroze 4 Blocks | 93.83% | 94.62% |

## 5. Conclusion

Reflecting on my plant leaf disease detection project, the biggest hurdle I faced was the domain gap between clean laboratory data and real world farming conditions. When I started with standard CNN architectures like EfficientNet, the models simply memorized the clean, uniform backgrounds of lab images. While they hit nearly 99% validation accuracy, they completely collapsed to around 25% when tested on real world field photos.

To solve this, I shifted my focus to instance segmentation. By progressing through different versions of the Segment Anything Model and landing on SAM 3, I found that mixing grayscale lab images from PlantVillage with cleanly segmented real world images from PlantDoc forced the CNNs to look at the actual leaf damage rather than background colors. This specific grayscale and segmented combination gave me my best CNN based result, stabilizing at an 85.26% testing accuracy across a complex 10 class mixed dataset.

However, the real breakthrough came when I transitioned to Vision Transformers. I discovered that by fine tuning these massive models, I didn't even need to manually crop the backgrounds anymore. Their self attention mechanisms natively figured out how to isolate the diseases.

Here is how the models peaked across different datasets:

- PlantVillage + PlantDoc (10 Classes):** The **DINOv2 Base** model (with 2 blocks unfrozen) was the absolute sweet spot. It achieved a fantastic **92.09%** testing accuracy on completely normal, unsegmented images, completely outperforming heavy SAM pipelines.
- Color Resilience (DINOv3 Large):** When I used the **DINOv3 Large** model at the PlantVillage + PlantDoc dataset, it hit **90.96%**. More impressively, because of its massive parameter count, it maintained a **90.40%** accuracy even when I completely stripped the color (Grayscale) from the images, proving it learns physical textures rather than superficial colors.
- Rice Disease Dataset (19 Classes):** The **DINOv2 Base** model (with 4 blocks unfrozen) delivered absolute highest performance of the entire project, dominating this massive dataset with a **94.62%** testing accuracy and near perfect recall across most fungal and bacterial classes.
- Wheat Plant Diseases (15 Classes):** The **DINOv2 Base** model generalized beautifully here as well, achieving a highly robust **91.73%** accuracy just by unfreezing 2 blocks.
- Potato + Tomato Dataset (10 Classes):** This was the only major bottleneck I hit. The **DINOv2 Base** model peaked at **70.67%** here. While it easily spotted healthy leaves and viral infections, it heavily struggled with cross species fungal overlap essentially confusing identical looking fungi, like Early Blight, across the two different plant species.